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COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGIES

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IS335 / CS498: Final Year Project Report – Mid of Semester One

Project Title: UDSM Service Desk System

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CHAPTER ONE

INTRODUCTION

1.1 General Introduction

A Service Desk System is a centralized platform or service that provides users, such as employees, students, or customers with a way to report issues, request support, and receive solutions from IT or non-IT departments (ITIL Foundation IT Service Management, 2019, AXELOS).

The UDSM Service Desk System is a system designed to make it easier for students, teachers, and staff to report and fix problems that face both students and staff. The service desk system works as a solution where everyone can ask for help and get updates about their issues quickly.

The system helps users to create tickets for their requests and receive updates in real time, it ensures that everyone stays informed and can track the progress of their issues without needing to rely on paper forms or waiting for answers.

For administrators and support teams, the system includes helpful tools like a calendar for scheduling, a directory of contacts, and options to assign tasks to the right people, also the system is built to handle lots of requests efficiently and simple to use.

This kind of system is especially important for the university and other clients of the university to use as it is secure, easy to use and a great user interface.

1.2 Statement of the problem

Universities face difficulties in managing support efficiently due to the lack of a centralized system, leading to delays, miscommunication, and resource mismanagement.

Students, clients and staff struggle to track issues, while support teams lack tools to prioritize and resolve tasks effectively.

Traditional manual methods are insufficient for handling high volumes of queries, such as storing a large pile of documents and tremendous number of emails, hence creating a need for a Service Desk System that improves communication, transparency

1.1 Objectives

1.1.1 Main objective

Development for the UDSM Service Desk System/ Help desk system.

1.1.2 Specific objectives

1. Make Communication Simple: Help users report problems and check updates easily through a user-friendly platform.
2. Solve Problems Faster: Ensure issues are fixed quickly by assigning tasks to the right people and keeping everyone updated.
3. Work More Efficiently: Provide tools like calendars and contact lists to help administrators and support teams organize and manage their work.
4. Be Clear and Accountable: Let users see the status of their requests so they know what's happening and who is handling their problem.
5. Handle Many Requests Easily: Ensure the system can manage lots of requests without becoming slow or hard to use.

1.2 Significance of the project

The Service Desk System is important because it solves several challenges faced by universities in managing support requests and improving operational efficiency. Its significance includes:

1. Improving Efficiency:

The system helps in quickly resolving problems and reducing delays in addressing issues, making the university's operations smoother and more efficient.

2. Single point of support:

By providing a single platform to manage both IT and non-IT issues, the system makes it easier for students, clients, faculty, and staff to get help and track the progress of their requests.

3. Enhanced Communication:

The system improves communication between users and support teams by providing real-time notifications, ticket tracking, and clear updates on the status of issues, helping everyone stay informed.

4. Better Resource Management:

Administrators can efficiently manage tasks, assign tickets to the right people, and track progress, ensuring resources are used effectively and no task is overlooked.

5. Scalability:

The system can handle a growing number of requests, making it suitable for large institutions like universities, where the volume of support requests can be high.

1.3 Project scope

This project focuses on creating a web-based service desk ticketing system tailored to the specific needs of UDSM. The system will provide a platform for staff and students to log issues related to university services, ensuring that these issues are efficiently tracked, managed, and resolved.

The system will include key features such as user authentication, ticket management, user role management, and time management functionalities.

Broadly, the main features of the system include:

1. User Authentication and Registration – The system should enable users to register with personal details such as name, email, phone number, registration/check number, and department. It should also provide secure login functionality with email and password, ensuring verification and approval before access.

Also, the admin can register users, support staff and technicians in the system.

In the side role management, it should support different user roles (admin, support, technician, and user), each with specific access levels, and different pages to be accessed for each of the roles.

2. Ticket Management – The next key feature is the ticket management functionality, it should allow users to create, view, and reply to tickets, Moreover, Admins, support staff and technicians should be able to assign tickets to technicians or other users.

A complete history of the ticket activities should also be shown for tracking and auditing.

Admins should be able to delete unassigned tickets and be able to update the ticket status (on whether it is open, in progress or closed).

3. Contact and Organizations Management – This section is important and will help the admins and other users to see other contacts within the system, and they should be able to assign the tickets to any relevant contact which they would think is suitable for this ticket.

It will enable the addition and management of contact details for relevant individuals and provide functionality to add and manage organizational details.

Target Users:

Primary Users: UDSM staff members and students who will raise and track their issues.

Secondary Users: Support staff, technicians, and administrators responsible for managing and resolving the issues.

1.4 Organization of the report

Chapter one: provides the introduction of the project title where we begin with the introduction of the problem, statement of the problem, main and specific objectives, project significance and finally the project scope.

Chapter two: This chapter consists of the literature review on existing solutions that match the intended solution that is to be made, also what goals that are supposed to be achieved on the intended system and the overview project plan of the intended system solution.

Chapter three: This chapter explains the details on the type of software development methodology that was being used in the project.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

Students and clients normally expect their issues to be solved in a short period of time, and they do not expect to stand in lines to wait for being served or their issues to be solved. Emails can be one way to solve this but think about 500 students sending emails about a problem in the university system, how would the university handle that amount?

That's where the service desk system solution arises to solve the problem faced by students and Clients, instead of waiting for days and hours on a long line to be served a minor issue about your ARIS system, why not issue a ticket to the specific department, and the ticket would be assigned to a specific staff to solve that problem for you?

The service desk system is a system that makes it easier for its user to assign tickets on several problems they are facing which are related to the university. The service desk system acts as a help desk to ensure all your problems are solved with pure transparency and in a short period of time.

How does service desk system improve efficiency?

A service desk serves as the backbone for IT support in organizations, enabling streamlined operations, effective communication, and enhanced user satisfaction. Here are the key ways in which a service desk improves efficiency:

1. Centralized Issue Resolution

Service desks collect all user questions and problems in one system. This makes it easier to track, organize, and fix issues quickly, saving time and resources. Reference: CompTIA (2023). The Role of Help Desks in IT Operations. Retrieved from CompTIA.org

2. Proactive Monitoring and Issue Prevention

Service desks use proactive monitoring to identify and resolve potential issues before they impact operations. Predictive analytics and monitoring tools ensure system reliability and prevent downtime. Reference: ISACA (2022). Leveraging Service Desks for Operational Resilience. Retrieved from ISACA.org

3. Performance Metrics and Continuous Improvement

By tracking metrics such as response times, resolution rates, and user satisfaction scores, service desks provide valuable insights for continuous improvement. This data-driven approach aligns support processes with organizational goals. Reference: CompTIA (2023). Service Desk Metrics for Improved IT Operations. Retrieved from CompTIA.org

Real-Time Notifications and User Experience in Support Systems

Real-time notifications have become an essential feature of modern support systems. They enable users to receive timely and relevant information about their activities, transactions, or interactions.

Notifications can be delivered through various methods, including email, SMS, push notifications, or in-app notifications.

Key Benefits of Real-Time Notifications

When users interact with a system, receiving timely notifications ensures they are always aware of what is happening. For example, in an e-commerce platform, when a user completes a purchase, receiving an order confirmation immediately after payment reassures them that their transaction was successful.

This smooth, seamless flow of communication eliminates uncertainty and contributes to a positive experience.

Without real-time updates, users may feel anxious, wondering whether their action, such as a

purchase or payment, was properly recorded.

Similarly, in other scenarios like appointment scheduling, instant confirmations ensure users are confident their appointments are set, removing the need for additional follow-ups.

Furthermore, real-time notifications effectively reduce support queries, saving time for both users and support teams. By proactively delivering important information, notifications eliminate the need for users to contact customer service to seek updates or clarifications.

2.1: Related Works

While planning to explain UDSM's journey to develop a Service Desk Portal System, let's first look at tools, frameworks, and best practices from around the world and Africa. We'll review globally recognized guidelines like ITIL, COBIT, and the ISO/IEC 20000 standard.

Then, we will focus on African approaches, including ideas from the AAU Toolkit and the World Bank's ICT in Higher Education Framework. This comparison will help UDSM gain the knowledge it needs to create a strong and effective Service Desk Portal that fits its specific needs.

Global Perspectives:

ITIL (Information Technology Infrastructure Library): ITIL is a widely adopted framework for IT service management (ITSM) that provides best practices for incident management, problem resolution, and service delivery. It is particularly useful for designing Service Desk systems that enhance user experience, optimize support workflows, and ensure service quality.

COBIT (Control Objectives for Information and Related Technologies): COBIT is a globally recognized framework for governance and management of enterprise IT. It ensures alignment between IT services and organizational goals, providing tools to evaluate and improve Service Desk performance.

ISO/IEC 20000: International Standard for IT Service Management: ISO/IEC 20000 is the international standard for ITSM, specifying requirements for designing, implementing, and managing effective IT service processes. It ensures consistency, efficiency, and continual improvement of systems such as Service Desk portals.

Service Desk Institute (SDI) Best Practice Standards: The SDI provides globally recognized standards and certifications for Service Desks, covering essential areas like customer satisfaction, incident response, knowledge management, and self-service capabilities.

African Perspectives:

The Association of African Universities (AAU) ICT Infrastructure Toolkit: The AAU Toolkit offers guidelines for implementing IT services in African universities, including strategies for Service Desk management. It highlights cost-effective solutions, capacity building, and the importance of local challenges like limited infrastructure and connectivity.

The World Bank's ICT in Higher Education: A Strategy for Africa: This report focuses on leveraging ICT to improve service delivery in African universities. It emphasizes the need for centralized IT systems, such as Service Desk portals, to enhance communication, support staff and students, and optimize resource management.

South African Higher Education ICT Policies: South African universities have implemented advanced ITSM frameworks, integrating Service Desk solutions to handle user queries, support academic operations, and streamline IT service delivery. These initiatives serve as key case studies for best practices within the region.

Tanzanian Perspectives:

Tanzania Commission for Universities (TCU) ICT Guidelines:

The TCU ICT guidelines promote the adoption of efficient IT systems to enhance administrative and academic processes. They encourage universities to integrate Service Desk solutions for improved support and user satisfaction.

University of Dar es Salaam (UDSM) Current ICT Strategy:

UDSM's existing ICT policies provide a foundation for implementing a Service Desk Portal. This strategy focuses on improving IT service delivery, incident management, and user engagement while aligning with global best practices and local needs.

2.1 : The Gap

The existing service desk systems are very helpful, but the UDSM Service Desk System fills important gaps by focusing on the university's specific needs. It allows users to create tickets easily and ensures tickets are assigned to the right admins, support staff, or technicians to solve

issues quickly.

The system combines information from different university departments to give a complete view of all requests, ensures smooth communication between users and support teams, and provides real-time updates on ticket progress. It also has a simple and easy-to-use design, making it accessible to everyone. This approach makes managing issues more efficient and helps UDSM improve its services for students, staff, and the whole university community.

CHAPTER THREE

METHODOLOGY

3.1 Introduction:

To make the service desk ticketing system for the University of Dar es Salaam (UDSM), we had to collaborate closely with UDSM staff to get the proper requirements and make the system work as intended, as at the end of the day, they will be the stakeholders themselves. To enable this, we chose the agile methodology to efficiently gather requirements, apply feedback and make the users involved continuously throughout the process.

3.2 Development Methodology:

We will adopt the Agile-Scrum methodology due to its iterative and adaptable nature, allowing us to respond to evolving user needs and gather and implement feedback.

Scrum Framework:

We will focus on dividing the development process into smaller, manageable units called sprints and ensure user feedback is collected and acted upon

The Scrum framework is a lightweight and adaptive methodology used to address complex problems and deliver high-value products iteratively. Scrum, rooted in Agile principles, organizes work into time-boxed iterations called **sprints**, typically lasting two to four weeks. Its focus is on transparency, inspection, and adaptation, allowing teams to respond to changes and incorporate stakeholder feedback effectively (Schwaber & Sutherland, 2020).

3.3 Requirement Analysis

Techniques for Collecting Requirements

To gather the requirements for the Service Desk System, our main requirement gathering method was interview. Kothari (2004) says that an interview is a powerful exchange of ideas, the answering of questions, and communication between two or more persons.

Overall, the following techniques were employed:

1. **Interviews:** Direct consultations with stakeholders, including staff at the University of Dar es Salaam (UDSM), to understand their needs, challenges, and expectations.

2. **Stakeholder Consultations:** Meetings with UDSM representatives to refine and validate the initial requirements, ensuring alignment with institutional goals.

System Design: Service Desk System

The design of the Service Desk System was meticulously planned and implemented using **Figma** for UI/UX design and various tools for development and architecture. The design process aimed to create a user-friendly, efficient, and scalable system for managing tickets and user interactions within the University of Dar es Salaam (UDSM).

3.4 Design Phases

1. Wireframes:

- Initial layouts were created using Figma to outline the structure of the application.
- Each wireframe included key components such as:
 - Sidebar for navigation.
 - Top bar for account management and logout functionality.
 - Main content area for ticket creation, management, and tracking.

2. Mockups and Prototypes:

- Figma was used to develop detailed mockups with visual styling, ensuring a consistent theme.
- Prototypes were built to simulate user interactions, helping gather feedback from stakeholders and ensuring the design met user expectations.

3. Revisions:

Feedback from stakeholders, including administrators, support staff, and end-users, was incorporated iteratively to improve usability and functionality

3.5 Implementation of the Service Desk System

The Service Desk System was implemented using modern tools and technologies to ensure it is user-friendly, efficient, responsive, and scalable. The frontend was developed using **React.js**, a

powerful framework that allows for the creation of a responsive and dynamic user interface. Navigation within the system is handled using **React Router DOM**, while **Styled Components** were utilized for modular and maintainable styling. To enhance the user experience, icons were provided using **React Icons** and **FontAwesome**.

The backend of the system was developed using **Node.js**, offering robust and scalable server-side functionality. For secure user authentication and access control, **JWT (JSON Web Tokens)** was implemented. The database used is **PostgreSQL**, a reliable and efficient relational database, which stores and manages critical data such as user roles (Admin, Support, End-User), tickets, logs, and notifications.

Several libraries and tools were integrated to enhance system functionality. **Axios** was used for seamless communication with the backend API. Version control and collaboration were managed using **Git** and **GitHub**, ensuring an organized development process with clearly tracked changes. Separate branches were maintained for feature development and bug fixes. Tools such as **Vite** were employed for fast and efficient builds, while **ESLint** was used to maintain clean, error-free code. During the design phase, **Figma** played a crucial role in creating and refining a responsive and user-friendly system interface. Security was a key consideration during implementation

Finally, the system was designed to be scalable and reliable, hosted on cloud infrastructure to support high availability and consistent performance.

3.6 Testing and Quality Assurance

- To ensure reliability, functionality, and user-friendliness of the UDSM Service Desk Ticketing System, we will conduct thorough tests.
- We will conduct unit testing to verify the functionality of individual components (e.g., user authentication, ticket creation).
- We will conduct integration testing to validate that different modules (e.g., User Management and Ticket Management) interact seamlessly. Tools such as Postman for API Testing can be used
- Lastly, we will conduct User Acceptance Testing (UAT), To ensure the system works as expected for UDSM staff and students. The approach will involve end-users in testing and

collecting feedback for final adjustments

3.7 Deployment

For the deployment of the UDSM Service Desk Ticketing System, we will look for the possibility of deploying the system on UDSM Servers, we can install all the necessary configurations and run it through that

This will involve collaborating with the university's IT team to allocate servers for hosting the system and ensuring the server meets the following requirements:

Operating System: Ubuntu Server will be preferred or other Linux-based OS.

RAM and CPU: Sufficient for handling concurrent requests based on user load.

Disk Space: Allocate adequate storage for application files and the PostgreSQL database.

Alternatively, we could look to host it in the cloud, which will offer scalability, flexibility, and reduced infrastructure management, however, might be a costly solution. We could utilise technologies such as AWS, or Google Cloud Platform to enable us to do this

3.8 Training and Documentation:

To ensure the effective use of the Service Desk Ticketing System, dedicated training sessions will be provided to both UDSM staff and students. These sessions will include:

- **Introductory Training:** Overview of the system's features, benefits, and user roles.
- **Hands-On Practice:** Interactive sessions where participants will create, assign, and resolve tickets in a simulated environment.
- **Role-Specific Training:**
 - For Users: Focus on raising tickets, tracking their status, and responding to updates.
 - For Support Staff and Technicians: Emphasis on managing assigned tickets, updating statuses, and maintaining ticket history.
 - For Administrators: Detailed instructions on managing user accounts, assigning roles, deleting tickets, and configuring system settings.

Comprehensive documentation will be prepared to guide users and administrators in operating the

system effectively. This includes:

1. User Guides:

- Step-by-step instructions on common tasks, such as logging in, raising tickets, and viewing ticket statuses.
- Illustrated examples and screenshots for ease of understanding.
- A troubleshooting section to address frequently encountered issues.

2. Administrator Manuals:

- Detailed guides for managing the system, including user verification, role assignments, ticket deletions, and system configurations.
- Guidelines for monitoring system performance and addressing technical issues.

We will also create the System Requirement Specification (SRS) Document and System Design Document (SDD) to familiarize stakeholders with the system.

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